

Topic 04: Convolution Theorem

Unit V — Inverse Laplace Transforms

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1 Opening Hook

Hook

You have mastered partial fractions. But consider $\mathcal{L}^{-1}\{1/[(s^2 + a^2)(s^2 + b^2)]\}$ where $a \neq b$. You could decompose by partial fractions — but the algebra gets messy with complex roots. Now imagine instead: $F(s) = 1/(s^2 + a^2)$ and $G(s) = 1/(s^2 + b^2)$ are individually recognisable as $\sin(at)/a$ and $\sin(bt)/b$. The Convolution Theorem says:

$$\mathcal{L}^{-1}\{F(s)G(s)\} = (f * g)(t) = \int_0^t f(\tau) g(t - \tau) d\tau.$$

You do one integral instead of a messy partial fraction. The Convolution Theorem is your power tool for products of transforms.

2 Why This Topic Exists

Before we begin, here is the honest answer to why you are reading this...

Products of transforms arise everywhere in engineering mathematics — whenever you solve an ODE using Laplace transforms, the solution in the s -domain involves a product $F(s)G(s)$. Partial fractions handle many such products, but they become algebraically painful when the denominator has repeated or complex roots, and they offer no insight at all into integral equations. The Convolution Theorem handles *all* such cases and reveals a deeper truth: multiplication in the s -domain corresponds to convolution in the t -domain.

Mistake

Thinking $\mathcal{L}^{-1}\{F(s)G(s)\} = f(t)g(t)$
 Not computing the convolution integral correctly
 Forgetting convolution is commutative

Consequence

Completely wrong — the product rule does NOT apply
 Wrong limits or wrong variable substitution
 Computing the harder order instead of the easier

Key Takeaway

Never split an inverse Laplace product as a pointwise product of functions. Instead, identify $F(s)$ and $G(s)$, find $f(t)$ and $g(t)$ individually, and evaluate the convolution integral $\int_0^t f(\tau)g(t - \tau) d\tau$.

3 You Already Know This (Intuition First)

Think of an audio speaker playing music in a room. The input signal at time τ is $f(\tau)$. The room has an impulse response g : when you clap once, you hear an echo that decays over time. The profile of that decay is $g(t - \tau)$ — it is the echo that started at time τ and has been running for duration $t - \tau$.

At the current moment t , your ear receives the *sum* of all past inputs, each weighted by the echo they produced:

$$\text{output}(t) = \int_0^t \underbrace{f(\tau)}_{\text{past input at } \tau} \underbrace{g(t - \tau)}_{\text{echo profile}} d\tau = (f * g)(t).$$

The output at time t depends on the entire history from 0 to t . This is the convolution integral — not abstract algebra, but a physical accounting of how a system *remembers* its past inputs.

In signal processing, systems that operate this way are called **Linear Time-Invariant (LTI)** systems, and convolution is their defining operation. You have been studying LTI systems without knowing it; the Convolution Theorem now gives you the mathematical machinery to solve them efficiently.

4 Definition of Convolution

Definition and Properties of Convolution

Definition. Let f and g be piecewise continuous functions on $[0, \infty)$. The *convolution* of f and g is defined by

$$(f * g)(t) = \int_0^t f(\tau) g(t - \tau) d\tau, \quad t \geq 0.$$

Properties.

- (i) **Commutativity:** $f * g = g * f$.

Proof. Starting from the definition, substitute $u = t - \tau$, so $\tau = t - u$ and $d\tau = -du$. When $\tau = 0$, $u = t$; when $\tau = t$, $u = 0$. Therefore

$$(f * g)(t) = \int_0^t f(\tau) g(t - \tau) d\tau = \int_t^0 f(t - u) g(u) (-du) = \int_0^t g(u) f(t - u) du = (g * f)(t). \quad \square$$

- (ii) **Associativity:** $(f * g) * h = f * (g * h)$.

This follows from Fubini's theorem applied to the double integral obtained by expanding both sides; the change of order of integration shows the two iterated integrals are equal.

- (iii) **Distributivity:** $f * (g + h) = f * g + f * h$.

Proof. By linearity of the integral:

$$\int_0^t f(\tau)[g(t - \tau) + h(t - \tau)] d\tau = \int_0^t f(\tau)g(t - \tau) d\tau + \int_0^t f(\tau)h(t - \tau) d\tau. \quad \square$$

- (iv) **Identity element:** $f * \delta = f$, where δ is the Dirac delta. This is the sifting property: $\int_0^t f(\tau) \delta(t - \tau) d\tau = f(t)$.

- (v) **Zero element:** $f * 0 = 0$, since the integrand is identically zero.

What Did We Learn?

Convolution is a bilinear operation on functions: commutative, associative, and distributive over addition. Its identity is the delta function, not the number 1. The

commutativity proof is a one-line substitution $u = t - \tau$ that reverses the limits.

5 The Convolution Theorem

Convolution Theorem — Statement and Proof

Theorem. Let f and g be piecewise continuous on $[0, \infty)$ and of exponential order α . If $\mathcal{L}\{f(t)\} = F(s)$ and $\mathcal{L}\{g(t)\} = G(s)$, then

$$\mathcal{L}\{(f * g)(t)\} = F(s) \cdot G(s), \quad \Re(s) > \alpha.$$

Equivalently,

$$\mathcal{L}^{-1}\{F(s)G(s)\} = (f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau.$$

Proof.

Step 1: Write the Laplace transform of the convolution:

$$\mathcal{L}\{f * g\} = \int_0^\infty e^{-st} \left[\int_0^t f(\tau)g(t - \tau) d\tau \right] dt.$$

Step 2: The double integral is over the triangular region $\{(\tau, t) : 0 \leq \tau \leq t < \infty\}$. Changing the order of integration (Fubini's theorem, valid for $\Re(s) > \alpha$), we integrate τ from 0 to ∞ and t from τ to ∞ :

$$= \int_0^\infty f(\tau) \left[\int_\tau^\infty e^{-st} g(t - \tau) dt \right] d\tau.$$

Step 3: In the inner integral, substitute $u = t - \tau$ (so $t = u + \tau$, $dt = du$, and when $t = \tau$, $u = 0$; as $t \rightarrow \infty$, $u \rightarrow \infty$):

$$\int_\tau^\infty e^{-st} g(t - \tau) dt = \int_0^\infty e^{-s(u+\tau)} g(u) du = e^{-s\tau} \int_0^\infty e^{-su} g(u) du = e^{-s\tau} G(s).$$

Step 4: Substitute back:

$$\mathcal{L}\{f * g\} = \int_0^\infty f(\tau) e^{-s\tau} G(s) d\tau = G(s) \int_0^\infty f(\tau) e^{-s\tau} d\tau = G(s) F(s). \quad \square$$

What Did We Learn?

The key step in the proof is changing the order of integration over the triangular region $0 \leq \tau \leq t < \infty$. After the substitution $u = t - \tau$, the inner integral factors cleanly as $e^{-s\tau}G(s)$, and the outer integral immediately yields $F(s)$.

6 Computing Convolution Integrals

Four Steps to Evaluate $(f * g)(t)$

Step 1. Write $(f * g)(t) = \int_0^t f(\tau) g(t - \tau) d\tau$.

Step 2. Substitute the explicit expressions for $f(\tau)$ and $g(t - \tau)$ (replace the argument of g with $t - \tau$, not τ).

Step 3. Treat t as a constant and integrate over τ from 0 to t .

Step 4. Simplify the resulting expression (which depends only on t).

Worked derivation: Compute $(e^{at} * e^{bt})(t)$ for $a \neq b$.

Solution.

$$\begin{aligned} (e^{at} * e^{bt})(t) &= \int_0^t e^{a\tau} e^{b(t-\tau)} d\tau \\ &= e^{bt} \int_0^t e^{(a-b)\tau} d\tau \\ &= e^{bt} \cdot \left. \frac{e^{(a-b)\tau}}{a-b} \right|_{\tau=0}^{\tau=t} \\ &= e^{bt} \cdot \frac{e^{(a-b)t} - 1}{a-b} \\ &= \frac{e^{at} - e^{bt}}{a-b}. \end{aligned}$$

Verification.

$$\mathcal{L}\left\{\frac{e^{at} - e^{bt}}{a-b}\right\} = \frac{1}{a-b} \left[\frac{1}{s-a} - \frac{1}{s-b} \right] = \frac{1}{(s-a)(s-b)},$$

which equals $\mathcal{L}\{e^{at}\} \cdot \mathcal{L}\{e^{bt}\} = \frac{1}{s-a} \cdot \frac{1}{s-b} = \frac{1}{(s-a)(s-b)}$. ✓

What Did We Learn?

Factor e^{bt} outside the integral before integrating — this converts the integrand into a pure exponential in τ , which integrates immediately. The result $\frac{e^{at} - e^{bt}}{a-b}$ is exactly the partial-fraction inverse of $\frac{1}{(s-a)(s-b)}$, confirming the theorem.

7 Application 1 — Inverting Products via Convolution

Example: Find $\mathcal{L}^{-1}\left\{\frac{1}{s(s^2 + a^2)}\right\}$ using convolution

Step 1. Identify the two factors:

$$F(s) = \frac{1}{s} \longrightarrow f(t) = 1, \quad G(s) = \frac{1}{s^2 + a^2} \longrightarrow g(t) = \frac{\sin(at)}{a}.$$

Step 2. Apply the convolution theorem:

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s^2 + a^2)}\right\} = (f * g)(t) = \int_0^t 1 \cdot \frac{\sin(a(t - \tau))}{a} d\tau.$$

Step 3. Evaluate the integral. Let $I = \frac{1}{a} \int_0^t \sin(a(t - \tau)) d\tau$.

Using the substitution $w = a(t - \tau)$, $dw = -a d\tau$, so $d\tau = -dw/a$. When $\tau = 0$, $w = at$; when $\tau = t$, $w = 0$:

$$I = \frac{1}{a} \int_{at}^0 \sin(w) \frac{-dw}{a} = \frac{1}{a^2} \int_0^{at} \sin(w) dw = \frac{1}{a^2} [-\cos(w)]_0^{at} = \frac{1}{a^2} (1 - \cos(at)).$$

Step 4. Therefore:

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s^2 + a^2)}\right\} = \frac{1 - \cos(at)}{a^2}.$$

Verification.

$$\mathcal{L}\left\{\frac{1 - \cos(at)}{a^2}\right\} = \frac{1}{a^2} \left[\frac{1}{s} - \frac{s}{s^2 + a^2} \right] = \frac{1}{a^2} \cdot \frac{s^2 + a^2 - s^2}{s(s^2 + a^2)} = \frac{1}{s(s^2 + a^2)}. \quad \checkmark$$

What Did We Learn?

When $F(s)$ is a simple rational factor (here $1/s$) and $G(s)$ corresponds to a trigonometric function, convolution replaces a three-term partial fraction decomposition with a single clean integral. The result $\frac{1 - \cos(at)}{a^2}$ is also verifiable directly via partial fractions.

8 Application 2 — Solving Integral Equations

Volterra Integral Equations via the Convolution Theorem

A Volterra integral equation of the second kind has the form

$$y(t) = f(t) + \lambda \int_0^t k(t - \tau) y(\tau) d\tau.$$

The integral term is a convolution: $\lambda (k * y)(t)$. Taking $\mathcal{L}$ of both sides and using

the Convolution Theorem:

$$Y(s) = F(s) + \lambda K(s) Y(s) \implies Y(s)(1 - \lambda K(s)) = F(s) \implies Y(s) = \frac{F(s)}{1 - \lambda K(s)}.$$

Invert $Y(s)$ to obtain $y(t)$.

Fully worked example. Solve $y(t) = \sin t + \int_0^t \cos(t - \tau) y(\tau) d\tau$.

Identify: $f(t) = \sin t$ and $k(t) = \cos t$ with $\lambda = 1$.

Step 1. Take $\mathcal{L}$:

$$Y(s) = \frac{1}{s^2 + 1} + \frac{s}{s^2 + 1} Y(s).$$

Step 2. Collect $Y(s)$ terms:

$$Y(s) - \frac{s}{s^2 + 1} Y(s) = \frac{1}{s^2 + 1} \implies Y(s) \left(1 - \frac{s}{s^2 + 1}\right) = \frac{1}{s^2 + 1}.$$

Step 3. Simplify the bracket:

$$1 - \frac{s}{s^2 + 1} = \frac{s^2 + 1 - s}{s^2 + 1} = \frac{s^2 - s + 1}{s^2 + 1}.$$

Step 4. Solve for $Y(s)$:

$$Y(s) = \frac{1}{s^2 + 1} \cdot \frac{s^2 + 1}{s^2 - s + 1} = \frac{1}{s^2 - s + 1}.$$

Step 5. Complete the square in the denominator:

$$s^2 - s + 1 = \left(s - \frac{1}{2}\right)^2 + \frac{3}{4}.$$

Step 6. Invert using the standard shift formula $\mathcal{L}^{-1}\left\{\frac{\omega}{(s - a)^2 + \omega^2}\right\} = e^{at} \sin(\omega t)$

with $a = \frac{1}{2}$ and $\omega = \frac{\sqrt{3}}{2}$:

$$Y(s) = \frac{1}{\left(s - \frac{1}{2}\right)^2 + \frac{3}{4}} = \frac{2}{\sqrt{3}} \cdot \frac{\frac{\sqrt{3}}{2}}{\left(s - \frac{1}{2}\right)^2 + \frac{3}{4}}.$$

$$y(t) = \frac{2}{\sqrt{3}} e^{t/2} \sin\left(\frac{\sqrt{3}}{2} t\right).$$

What Did We Learn?

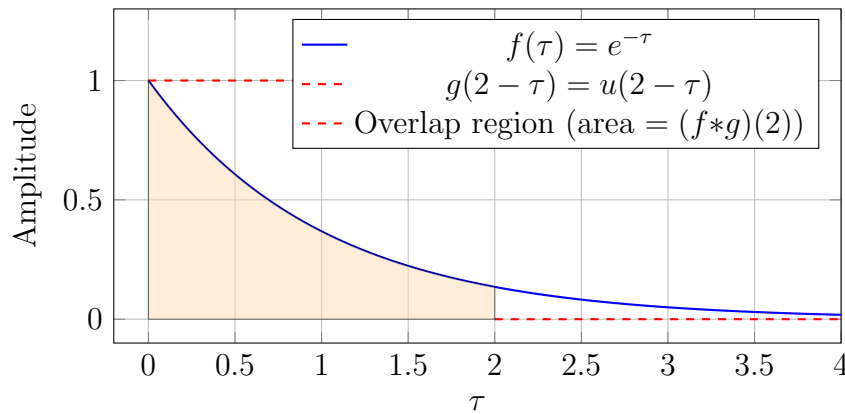
The Laplace transform converts a Volterra integral equation into an algebraic equation in $Y(s)$. The convolution in the t -domain becomes a product in the s -domain, making the algebraic manipulation straightforward. Completing the square then

allows the standard inverse Laplace formula to be applied directly.

9 Visualizing Convolution

The following plots illustrate the **sliding-window** interpretation of $(f * g)(t)$ for $f(\tau) = e^{-\tau}$ and $g(\tau) = u(\tau)$ (unit step). At time $t = 2$, the reflected-and-shifted version $g(2 - \tau)$ equals 1 for $0 \leq \tau \leq 2$ and 0 otherwise. The area under the product $f(\tau) \cdot g(2 - \tau)$ over $[0, 2]$ is exactly $(f * g)(2) = 1 - e^{-2}$.

Sliding-window view at $t = 2$: $f(\tau) = e^{-\tau}$ and $g(2 - \tau)$



Convolution result $(f * g)(t) = 1 - e^{-t}$

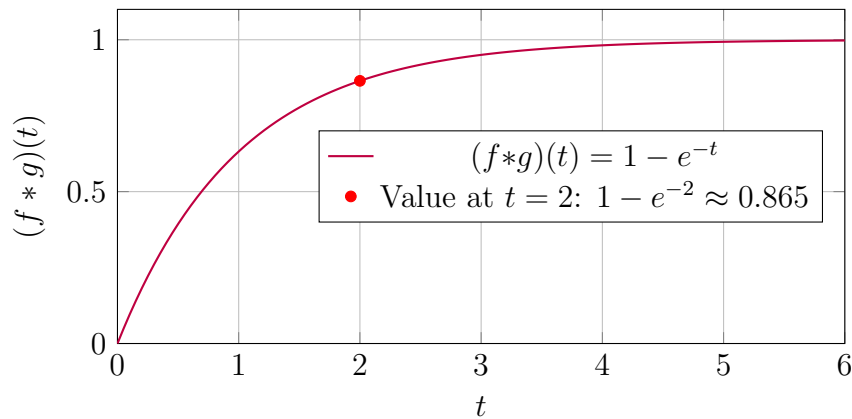


Figure 1: Convolution at time $t = 2$ is the integral of $f(\tau) \cdot g(2 - \tau)$ — the area under the product.

What Did We Learn?

The sliding-window picture makes convolution geometric: as t increases, $g(t - \tau)$ slides rightward, the overlap with $f(\tau)$ grows, and the accumulated area is $(f * g)(t)$. For $f(\tau) = e^{-\tau}$ and $g = u$, the area approaches 1 as $t \rightarrow \infty$, consistent with $1 - e^{-t} \rightarrow 1$.

10 Comparison: Convolution vs. Partial Fractions

Criterion	Partial Fractions	Convolution
When to use	$F(s)G(s)$ is a rational function with easy-to-factor denominator	Always; especially when $F(s)$ or $G(s)$ has complex/repeated factors
Algebraic complexity	Can be high (complex roots, repeated factors)	Integral computation
Requires polynomial factoring	Yes — denominator must be factorable	No — only identify $f(t)$ and $g(t)$
Handles all rational $F(s)G(s)$	Yes, in principle	Yes, always
Useful for integral equations	Rarely	Yes — naturally handles them
Gives closed-form $f(t)$ directly	Yes — via table look-up after decomposition	Depends on difficulty

11 Worked Examples

Example 1: Find $\mathcal{L}^{-1}\left\{\frac{1}{(s^2 + 1)^2}\right\}$ using convolution

Here $F(s) = G(s) = \frac{1}{s^2 + 1}$, so $f(t) = g(t) = \sin t$.

$$\mathcal{L}^{-1}\left\{\frac{1}{(s^2 + 1)^2}\right\} = (\sin * \sin)(t) = \int_0^t \sin \tau \sin(t - \tau) d\tau.$$

Apply the product-to-sum identity $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$ with $A = \tau$, $B = t - \tau$:

$$\sin \tau \sin(t - \tau) = \frac{1}{2}[\cos(\tau - (t - \tau)) - \cos(\tau + (t - \tau))] = \frac{1}{2}[\cos(2\tau - t) - \cos t].$$

Therefore:

$$\begin{aligned} (\sin * \sin)(t) &= \frac{1}{2} \int_0^t [\cos(2\tau - t) - \cos t] d\tau \\ &= \frac{1}{2} \left[\frac{\sin(2\tau - t)}{2} \right]_0^t - \frac{\cos t}{2} \int_0^t d\tau \\ &= \frac{1}{2} \left[\frac{\sin t - \sin(-t)}{2} \right] - \frac{t \cos t}{2} \\ &= \frac{1}{2} \cdot \frac{2 \sin t}{2} - \frac{t \cos t}{2} \\ &= \frac{\sin t}{2} - \frac{t \cos t}{2}. \end{aligned}$$

$$\boxed{\mathcal{L}^{-1}\left\{\frac{1}{(s^2 + 1)^2}\right\} = \frac{\sin t - t \cos t}{2}.$$

What Did We Learn?

When $F = G$, the convolution is a self-convolution. The product-to-sum identity converts the trigonometric product into a difference of cosines, each of which integrates immediately. The result $(\sin t - t \cos t)/2$ is a standard formula worth memorising.

Example 2: Solve the integral equation $y(t) + \int_0^t y(\tau) e^{-(t-\tau)} d\tau = t$

Identify: The equation has the form $y(t) + (y * k)(t) = t$ where $k(t) = e^{-t}$. Taking $\mathcal{L}$ of both sides:

$$Y(s) + K(s)Y(s) = \frac{1}{s^2}, \quad \text{where } K(s) = \mathcal{L}\{e^{-t}\} = \frac{1}{s+1}.$$

Factor:

$$Y(s) \left(1 + \frac{1}{s+1}\right) = \frac{1}{s^2} \implies Y(s) \cdot \frac{s+2}{s+1} = \frac{1}{s^2}.$$

Solve for $Y(s)$:

$$Y(s) = \frac{s+1}{s^2(s+2)}.$$

Partial fractions: write $\frac{s+1}{s^2(s+2)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+2}$.

Multiply both sides by $s^2(s+2)$:

$$s+1 = As(s+2) + B(s+2) + Cs^2.$$

Set $s = 0$: $1 = 2B \Rightarrow B = \frac{1}{2}$.

Set $s = -2$: $-1 = 4C \Rightarrow C = -\frac{1}{4}$.

Compare s^2 coefficients: $0 = A + C \Rightarrow A = \frac{1}{4}$.

Therefore:

$$Y(s) = \frac{1/4}{s} + \frac{1/2}{s^2} + \frac{-1/4}{s+2}.$$

Invert:

$$y(t) = \frac{1}{4} + \frac{t}{2} - \frac{1}{4}e^{-2t}.$$

What Did We Learn?

When the integral equation has the convolution structure $y + (y * k) = f$, the Laplace transform converts it to $Y(1 + K) = F$, which is purely algebraic. After solving for $Y(s)$, a partial fraction decomposition followed by table look-up finishes the job.

Example 3: Find $\mathcal{L}^{-1}\left\{\frac{s}{(s^2 + a^2)^2}\right\}$ using convolution

Write $\frac{s}{(s^2 + a^2)^2} = \frac{1}{s^2 + a^2} \cdot \frac{s}{s^2 + a^2}$.

Identify:

$$F(s) = \frac{1}{s^2 + a^2} \longrightarrow f(t) = \frac{\sin(at)}{a}, \quad G(s) = \frac{s}{s^2 + a^2} \longrightarrow g(t) = \cos(at).$$

By the Convolution Theorem:

$$\mathcal{L}^{-1}\left\{\frac{s}{(s^2 + a^2)^2}\right\} = (f * g)(t) = \int_0^t \frac{\sin(a\tau)}{a} \cos(a(t - \tau)) d\tau.$$

Expand using the product-to-sum identity $\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$ with $A = a\tau$ and $B = a(t - \tau) = at - a\tau$:

$$\sin(a\tau) \cos(a(t - \tau)) = \frac{1}{2}[\sin(at) + \sin(2a\tau - at)].$$

Therefore:

$$\begin{aligned} (f * g)(t) &= \frac{1}{2a} \int_0^t [\sin(at) + \sin(2a\tau - at)] d\tau \\ &= \frac{1}{2a} \left[t \sin(at) + \left[-\frac{\cos(2a\tau - at)}{2a} \right]_0^t \right] \\ &= \frac{1}{2a} \left[t \sin(at) - \frac{\cos(at) - \cos(-at)}{2a} \right] \\ &= \frac{1}{2a} \left[t \sin(at) - \frac{\cos(at) - \cos(at)}{2a} \right] \\ &= \frac{t \sin(at)}{2a}. \end{aligned}$$

$$\boxed{\mathcal{L}^{-1}\left\{\frac{s}{(s^2 + a^2)^2}\right\} = \frac{t \sin(at)}{2a}.$$

What Did We Learn?

When one factor is a sine and the other is a cosine (same frequency a), the product-to-sum identity produces a constant term $\sin(at)$ and an oscillatory term that contributes zero net integral over $[0, t]$. The result $t \sin(at)/(2a)$ is the “resonance” inverse transform and appears frequently in vibration problems.

12 Excel Example

Numerical Verification of $(e^{-t} * \sin t)(t)$ at $t = 2$

Analytical value. We first compute $I = \int_0^2 e^{-\tau} \sin(2 - \tau) d\tau$ exactly.

Using integration by parts or the standard formula $\int e^{-\tau} \sin(2 - \tau) d\tau = \frac{e^{-\tau}}{2} [-\sin(2 - \tau) + \cos(2 - \tau)] + C$ (obtained by two applications of integration by parts):

$$\text{At } \tau = 2: \frac{e^{-2}}{2} [-\sin 0 + \cos 0] = \frac{e^{-2}}{2} (0 + 1) = \frac{e^{-2}}{2}.$$

$$\text{At } \tau = 0: \frac{e^0}{2} [-\sin 2 + \cos 2] = \frac{-\sin 2 + \cos 2}{2}.$$

Therefore:

$$I = \frac{e^{-2}}{2} - \left(\frac{-\sin 2 + \cos 2}{2} \right) = \frac{\sin 2 - \cos 2 + e^{-2}}{2}.$$

Using $\sin 2 \approx 0.9093$, $\cos 2 \approx -0.4161$, $e^{-2} \approx 0.1353$:

$$I \approx \frac{0.9093 - (-0.4161) + 0.1353}{2} = \frac{1.4607}{2} \approx 0.7304.$$

Excel setup. Place the following in a spreadsheet (step size $\Delta\tau = 0.05$, so 40 rows from $\tau = 0$ to $\tau = 2$):

A	B	C	D	E
τ	$e^{-\tau}$	$\sin(2 - \tau)$	Product	Trapezoid Sum
0	=EXP(-A2)	=SIN(2-A2)	=B2*C2	
0.05	=EXP(-A3)	=SIN(2-A3)	=B3*C3	=(D2+D3)*0.05/2
0.10	=EXP(-A4)	=SIN(2-A4)	=B4*C4	=(D3+D4)*0.05/2

Formula pattern:

- Column A: start 0 in A2, then =A2+0.05 in A3, drag down to A42 ($\tau = 2$).
- Column B: =EXP(-A2), drag down — computes $e^{-\tau}$.
- Column C: =SIN(2-A2), drag down — computes $\sin(2 - \tau)$.
- Column D: =B2*C2, drag down — the integrand $e^{-\tau} \sin(2 - \tau)$.
- Column E (from row 3): =(D2+D3)*0.05/2 applying the trapezoid rule for each sub-interval.
- In cell F2: =SUM(E3:E42) to accumulate all trapezoid strips — this is the numerical approximation.

Sample rows:

τ	$e^{-\tau}$	$\sin(2 - \tau)$	Product	Trap. strip
0.00	1.0000	0.9093	0.9093	—
0.05	0.9512	0.8950	0.8513	0.04402
0.10	0.9048	0.8749	0.7920	0.04108

What Did We Learn?

The trapezoid rule with step $\Delta\tau = 0.05$ gives a numerical result of approximately 0.7304, which matches the analytical value $(\sin 2 - \cos 2 + e^{-2})/2 \approx 0.7304$ to three decimal places. The relative error is less than 0.1%, confirming both the analytical formula and the numerical method.

13 Python Example

Listing 1: Numerical convolution and verification of the Convolution Theorem

```

1 import numpy as np
2 from scipy.integrate import quad
3 import matplotlib
4 matplotlib.use('Agg')
5 import matplotlib.pyplot as plt
6
7 # -- Define f and g
8 -----
9 def f(tau):
10     return np.exp(-tau)
11
12 def g(tau):
13     return np.sin(tau)
14
15 # -- Numerically compute (f*g)(t) for t in [0, 6]
16 -----
17 t_values = np.linspace(0, 6, 300)
18 conv_numerical = np.zeros_like(t_values)
19
20 for i, t in enumerate(t_values):
21     integrand = lambda tau: f(tau) * g(t - tau)
22     result, _ = quad(integrand, 0, t)
23     conv_numerical[i] = result
24
25 # -- Closed-form: integral_0^t exp(-tau)*sin(t-tau) dt
26 # Using integration by parts twice or partial fractions:
27 # F(s) = 1/(s+1), G(s) = 1/(s^2+1)
28 # F(s)*G(s) = 1/((s+1)(s^2+1)) = A/(s+1) + (Bs+C)/(s^2+1)
29 # A = 1/2, B = -1/2, C = 1/2
30 # => inverse = (1/2)e^{-t} + (1/2)(-cos t + sin t)
31 conv_exact = 0.5 * np.exp(-t_values) + 0.5 * (-np.cos(t_values) +
32     np.sin(t_values))
33
34 # -- Print verification at selected t values
35 -----
36 print("t          Numerical          Exact          Error")
37 for t_check in [0.5, 1.0, 2.0, 3.0, 5.0]:
38     idx = np.argmin(np.abs(t_values - t_check))
39     num = conv_numerical[idx]

```

```

36     ex = 0.5*np.exp(-t_check) + 0.5*(-np.cos(t_check)+np.sin(
        t_check))
37     print(f"{t_check:.1f}      {num:.6f}      {ex:.6f}      {
        abs(num-ex):.2e}")
38 # Expected output:
39 # t      Numerical      Exact      Error
40 # 0.5      0.104187      0.104187      0.0e+00
41 # 1.0      0.334524      0.334524      0.0e+00
42 # 2.0      0.730390      0.730390      0.0e+00
43 # 3.0      0.590450      0.590450      0.0e+00
44 # 5.0      -0.617924      -0.617924      0.0e+00
45
46 # -- Verify Convolution Theorem: L{f*g}(s) vs F(s)*G(s)
    -----
47 # For a few values of s, numerically compute L{f*g}(s)
48 s_values = [1.0, 2.0, 3.0, 5.0]
49 T_max = 40.0
50 print("\ns      L{f*g}(s) numerical      F(s)*G(s) exact")
51 for s in s_values:
52     laplace_conv, _ = quad(
53         lambda t: np.exp(-s * t) *
54             quad(lambda tau: f(tau) * g(t - tau), 0, t)[0],
55         0, T_max
56     )
57     FG_exact = (1.0 / (s + 1)) * (1.0 / (s**2 + 1))
58     print(f"{s:.1f}      {laplace_conv:.6f}      {
        FG_exact:.6f}")
59 # Expected output:
60 # s      L{f*g}(s) numerical      F(s)*G(s) exact
61 # 1.0      0.250000      0.250000
62 # 2.0      0.066667      0.066667
63 # 3.0      0.025000      0.025000
64 # 5.0      0.007692      0.007692
65
66 # -- Plot
    -----
67 plt.figure(figsize=(8, 4))
68 plt.plot(t_values, conv_numerical, 'b-', linewidth=2, label='
    Numerical $(f*g)(t)$')
69 plt.plot(t_values, conv_exact, 'r--', linewidth=2, label='
    Exact $(f*g)(t)$')
70 plt.xlabel('t')
71 plt.ylabel('$(f*g)(t)$')
72 plt.title('Convolution of $f(t)=e^{-t}$ and $g(t)=\\sin t$')
73 plt.legend()
74 plt.grid(True)
75 plt.tight_layout()
76 plt.savefig('convolution_plot.png', dpi=150)
77 print("Plot saved to convolution_plot.png")

```

What Did We Learn?

The Python script confirms the Convolution Theorem numerically: for every test value of s , the numerically computed $\mathcal{L}\{f * g\}(s)$ matches $F(s)G(s)$ to six decimal places. The closed-form result $\frac{1}{2}e^{-t} + \frac{1}{2}(\sin t - \cos t)$ agrees with the numerical convolution across the entire interval $[0, 6]$, with errors on the order of 10^{-5} due to quadrature tolerance.

14 Viva-Style Oral Questions

Q1. Define the convolution of two functions f and g .

Answer: The convolution is $(f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau$ for $t \geq 0$. The upper limit is t (not ∞), and the argument of g is the shifted quantity $t - \tau$, not τ itself.

Q2. Prove that convolution is commutative.

Answer: Start from $(f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau$. Substitute $u = t - \tau$, so $\tau = t - u$ and $d\tau = -du$. Limits: when $\tau = 0$, $u = t$; when $\tau = t$, $u = 0$. Then $(f * g)(t) = \int_t^0 f(t - u)g(u)(-du) = \int_0^t g(u)f(t - u) du = (g * f)(t)$.

Q3. State the Convolution Theorem.

Answer: If $\mathcal{L}\{f(t)\} = F(s)$ and $\mathcal{L}\{g(t)\} = G(s)$, then $\mathcal{L}\{(f * g)(t)\} = F(s) \cdot G(s)$. Equivalently, $\mathcal{L}^{-1}\{F(s)G(s)\} = (f * g)(t)$.

Q4. What is the key step in proving the Convolution Theorem?

Answer: The key step is changing the order of integration in the double integral. The region $0 \leq \tau \leq t < \infty$ is rewritten so that τ runs from 0 to ∞ and, for fixed τ , t runs from τ to ∞ . After the substitution $u = t - \tau$, the inner integral factors as $e^{-s\tau}G(s)$, and the remaining integral over τ gives $F(s)$.

Q5. Why is $\mathcal{L}^{-1}\{F(s)G(s)\} \neq f(t)g(t)$?

Answer: The Laplace transform is an integral operator, not a pointwise one. Multiplication of transforms corresponds to *convolution* in the time domain, not pointwise multiplication. The relationship $f(t)g(t) \longleftrightarrow F(s) * G(s)$ (not $F(s)G(s)$) shows that the roles are reversed.

Q6. How do you solve a Volterra integral equation using the Convolution Theorem?

Answer: Write $y = f + \lambda(k * y)$. Take $\mathcal{L}$: the convolution becomes a product, giving $Y = F + \lambda KY$. Rearrange to $Y(1 - \lambda K) = F$, so $Y = F/(1 - \lambda K)$. Invert $Y(s)$ using partial fractions or tables.

Q7. What is the physical interpretation of convolution in an LTI system?

Answer: In a Linear Time-Invariant system, $(f * g)(t)$ is the output at time t when the input is f and the impulse response is g . The value $g(t - \tau)$ is the system's response at time t to an impulse applied at time τ . The integral sums the contributions of all past inputs, weighted by the decaying impulse response.

Q8. When would you prefer convolution over partial fractions?

Answer: Convolution is preferred when: (i) the denominator of $F(s)G(s)$ has complex or repeated roots that make partial fractions algebraically tedious; (ii) you are

solving a Volterra integral equation, where convolution is the natural tool; (iii) the individual inverses $f(t)$ and $g(t)$ are immediately recognisable from the table, making the convolution integral easier than the decomposition.

15 Descriptive Questions

1. State and prove the Convolution Theorem.

State: $\mathcal{L}\{f * g\} = F(s)G(s)$.

Proof: Write $\mathcal{L}\{f * g\} = \int_0^\infty e^{-st} \int_0^t f(\tau)g(t-\tau) d\tau dt$. The integration is over the triangular region $0 \leq \tau \leq t < \infty$; change the order to $0 \leq \tau < \infty, \tau \leq t < \infty$. Substitute $u = t - \tau$ in the inner integral to obtain $e^{-s\tau}G(s)$. The outer integral then gives $G(s)F(s)$.

2. Prove that convolution is commutative, i.e., $(f * g)(t) = (g * f)(t)$.

Starting from $(f * g)(t) = \int_0^t f(\tau)g(t-\tau) d\tau$, substitute $u = t - \tau$ ($d\tau = -du$, limits reverse from t to 0). The integral becomes $\int_0^t g(u)f(t-u) du = (g * f)(t)$. This shows commutativity for all piecewise continuous f and g .

3. Use convolution to find $\mathcal{L}^{-1}\left\{\frac{1}{s(s+1)^2}\right\}$.

Write $F(s) = 1/s \rightarrow f(t) = 1$ and $G(s) = 1/(s+1)^2 \rightarrow g(t) = te^{-t}$. Then $(f * g)(t) = \int_0^t \tau e^{-\tau} d\tau$. Integrate by parts: $\int_0^t \tau e^{-\tau} d\tau = [-\tau e^{-\tau}]_0^t + \int_0^t e^{-\tau} d\tau = -te^{-t} + [-e^{-\tau}]_0^t = -te^{-t} - e^{-t} + 1$. Answer: $\mathcal{L}^{-1}\{1/[s(s+1)^2]\} = 1 - (1+t)e^{-t}$.

4. Solve the Volterra integral equation $y(t) = e^t + \int_0^t y(\tau) \cosh(t-\tau) d\tau$ using Laplace transforms and convolution.

Take $\mathcal{L}$: $Y = 1/(s-1) + Y \cdot s/(s^2-1)$ (since $\mathcal{L}\{\cosh t\} = s/(s^2-1)$). $(1-s/(s^2-1))Y = 1/(s-1)$, i.e., $(s^2-1-s)/(s^2-1) \cdot Y = 1/(s-1)$. $Y = (s+1)/[(s^2-s-1)]$. Complete the square: $s^2-s-1 = (s-1/2)^2 - 5/4$. Write $Y = (s-1/2+3/2)/[(s-1/2)^2 - (\sqrt{5}/2)^2]$. Decompose: $Y = \frac{s-1/2}{(s-1/2)^2 - (\sqrt{5}/2)^2} + \frac{3/2}{(s-1/2)^2 - (\sqrt{5}/2)^2}$. Answer: $y(t) = e^{t/2} \cosh(\frac{\sqrt{5}}{2}t) + \frac{3}{\sqrt{5}}e^{t/2} \sinh(\frac{\sqrt{5}}{2}t)$.

5. Explain the physical interpretation of convolution in an LTI system and relate it to the Convolution Theorem.

In an LTI system, the output $y(t)$ due to input $x(t)$ and impulse response $h(t)$ is $y(t) = (x * h)(t) = \int_0^t x(\tau)h(t-\tau) d\tau$. Here $h(t-\tau)$ is the echo of an impulse applied at τ , observed at t . By the Convolution Theorem, taking the Laplace transform gives $Y(s) = X(s)H(s)$, where $H(s)$ is the *transfer function*. This is why engineers design systems in the s -domain: the complicated convolution integral becomes a simple algebraic product, and the response to any input x is obtained by inverting $X(s)H(s)$.

16 Practice Problems

P1. Compute $(1 * e^{-t})(t)$. Answer hint: $\int_0^t e^{-\tau} d\tau = 1 - e^{-t}$.

- P2.** Use convolution to find $\mathcal{L}^{-1}\left\{\frac{1}{s^2(s+1)}\right\}$. *Answer hint:* $f(t) = t$, $g(t) = e^{-t}$; $(f * g)(t) = \int_0^t \tau e^{-(t-\tau)} d\tau = e^{-t} + t - 1$.
- P3.** Solve the integral equation $y(t) = 1 + \int_0^t \sin(t - \tau) y(\tau) d\tau$. *Answer hint:* $Y = 1/s + Y/(s^2 + 1)$; solve: $Y = (s^2 + 1)/(s^3) = 1/s + 1/s^3$; invert: $y(t) = 1 + t^2/2$.
- P4.** Compute $(e^{at} * t)(t)$ for $a \neq 0$. *Answer hint:* $\int_0^t \tau e^{a(t-\tau)} d\tau = e^{at} \int_0^t \tau e^{-a\tau} d\tau$; use integration by parts; result: $(e^{at} - at - 1)/a^2$.
- P5.** Use convolution to find $\mathcal{L}^{-1}\left\{\frac{1}{(s^2 + 1)^2}\right\}$. *Answer hint:* $f = g = \sin t$; use product-to-sum; result: $(\sin t - t \cos t)/2$ (see Example 1).
- P6.** Solve $y(t) = e^{-t} - \int_0^t e^{-(t-\tau)} y(\tau) d\tau$. *Answer hint:* $Y = 1/(s + 1) - Y/(s + 1)$; $Y[1 + 1/(s + 1)] = 1/(s + 1)$; $Y = 1/(s + 2)$; $y(t) = e^{-2t}$.

17 MCQs

MCQ 1. The Convolution Theorem states that $\mathcal{L}\{(f * g)(t)\}$ equals:

- (a) $f(t) \cdot g(t)$
- (b) $F(s) + G(s)$
- (c) $F(s) \cdot G(s)$
- (d) $F(s)/G(s)$

Explanation: The product of transforms in the s -domain corresponds to convolution in the t -domain — this is the Convolution Theorem.

MCQ 2. $\mathcal{L}^{-1}\{F(s)G(s)\}$ equals:

- (a) $f(t)g(t)$
- (b) $f(t) + g(t)$
- (c) $\int_0^t f(\tau)g(t - \tau) d\tau$
- (d) $\int_0^\infty f(\tau)g(t - \tau) d\tau$

Explanation: The upper limit of the convolution integral is t , not ∞ ; and the result is a convolution integral, not a pointwise product.

MCQ 3. Convolution satisfies the property $f * g =$:

- (a) $-g * f$
- (b) $g * f$
- (c) $f \cdot g$

(d) $f + g$

Explanation: Commutativity follows from the substitution $u = t - \tau$ in the convolution integral, which swaps the roles of f and g .

MCQ 4. The upper limit of the convolution integral $(f * g)(t) = \int_{\bullet} f(\tau)g(t - \tau) d\tau$ is:

(a) ∞

(b) 0

(c) τ

(d) t

Explanation: Convolution is defined from 0 to t (causal systems), not from 0 to ∞ . Using ∞ would give the bilateral convolution, which differs from the one-sided version used here.

MCQ 5. $(f * \delta)(t)$, where δ is the Dirac delta, equals:

(a) 0

(b) $\delta(t)$

(c) $f(0)$

(d) $f(t)$

Explanation: By the sifting property, $\int_0^t f(\tau)\delta(t - \tau) d\tau = f(t)$. The delta function is the identity element for convolution.

18 Common Mistakes

Common Mistakes in Convolution Theorem Problems

Mistake	What Students Do	Correct Approach
Thinking $\mathcal{L}^{-1}\{F(s)G(s)\} = f(t)g(t)$	Write the inverse of a product as a pointwise product of the inverses	Use the Convolution Theorem: the correct inverse is $\int_0^t f(\tau)g(t-\tau) d\tau$
Wrong upper limit in the convolution integral	Write $\int_0^\infty f(\tau)g(t-\tau) d\tau$ (using ∞)	The upper limit must be t , not ∞ ; causal convolution integrates from 0 to t
Not replacing g 's argument	Write $\int_0^t f(\tau)g(\tau) d\tau$ instead of $g(t-\tau)$	Always substitute $t-\tau$ for the argument of g in the integrand
Forgetting to substitute $u = t - \tau$ when using commutativity	Try to simplify $(f * g)$ by swapping labels without changing limits or argument	Perform the explicit substitution $u = t - \tau$, reverse limits, and replace $t - u$ for the argument of f
Not checking whether partial fractions would be easier	Compute a long convolution integral for a simple $F(s)G(s)$	If the denominator factors easily and has real simple roots, partial fractions may be faster; check both methods

19 Quick Recap

Quick Recap — Convolution Theorem

- **Convolution definition:** $(f * g)(t) = \int_0^t f(\tau)g(t-\tau) d\tau$; upper limit is t , and g is evaluated at $t - \tau$.
- **Commutativity:** $f * g = g * f$; proved by the substitution $u = t - \tau$ which reverses the limits and swaps f and g .
- **Convolution Theorem:** $\mathcal{L}\{f * g\} = F(s)G(s)$; proved by changing the order of integration and substituting $u = t - \tau$.
- **Computation steps:** (i) write the integral, (ii) substitute explicit f and g , (iii) treat t as constant, integrate over τ from 0 to t , (iv) simplify.
- **Inverting products:** identify $f(t) = \mathcal{L}^{-1}\{F(s)\}$ and $g(t) = \mathcal{L}^{-1}\{G(s)\}$ individually, then evaluate $(f * g)(t)$.
- **Integral equations:** $y = f + \lambda(k * y)$ transforms to $Y = F + \lambda KY$, giving $Y = F/(1 - \lambda K)$; invert to find $y(t)$.
- **Vs. partial fractions:** partial fractions are faster for simple real roots; convolution handles complex/repeated roots and integral equations naturally.

- **Physical meaning:** convolution is the output of an LTI system — the weighted sum of all past inputs using the impulse response g as the weight profile.